

PROJECT PROPOSAL

Project Title

Smart Grocery and Budget Calculator with Quantity and Price Sorting

Problem Statement

Grocery shopping can be difficult to organize when people have many items to buy. It can also be hard to calculate the total cost when buying different quantities of items and to identify which items are the most expensive. This project aims to create a simple Python program that helps users organize their grocery list and keep track of their expenses.

Project Objectives

- To create a program that allows users to add grocery items.
- To calculate the total price of each grocery item based on its price and quantity.
- To calculate the overall cost of the grocery list.
- To organize grocery items from the highest to the lowest price.
- To help users keep track of their grocery expenses.
- To practice using Python lists, conditions, loops, and the math library.

Planned Features

- Add Item – Allows the user to enter an item name, price, and quantity.
- View Grocery List – Displays all added grocery items and their information.
- Compute Total Cost – Calculates the total cost of all grocery items.
- Sort Items by Price – Arranges items from the highest to the lowest price.
- Menu System – Allows users to choose different actions.
- Exit – Ends the program when the user is finished.
- Automatic Calculations – Calculates each item's total using price $\times$ quantity.
- Budget Tracking – Shows the user how much they are spending.

Planned Inputs and Outputs

Inputs

- Item Name – String
- Item Price – Float
- Quantity – Integer
- Menu Choice – Integer

Outputs

- Grocery item names
- Item prices
- Quantities
- Total price for each item
- Overall total cost
- Grocery items sorted from highest to lowest price
- Program messages such as item added successfully and program ended

LOGIC PLAN

START

IMPORT math

CREATE empty list item_names

CREATE empty list item_prices

CREATE empty list item_quantities

SET choice = 0

WHILE choice is not 5

 DISPLAY "=====

 DISPLAY "SMART GROCERY LIST & BUDGET"

 DISPLAY "1. Add Item"

 DISPLAY "2. View Grocery List"

 DISPLAY "3. Compute Total Cost"

 DISPLAY "4. Sort Items by Price"

```
DISPLAY "5. Exit"
DISPLAY "=====
```

```
INPUT choice
```

```
IF choice = 1 THEN
```

```
    INPUT item name
    INPUT item price
    INPUT quantity
```

```
    ADD item name to item_names
    ADD item price to item_prices
    ADD quantity to item_quantities
```

```
    DISPLAY "Item added successfully!"
```

```
ELSE IF choice = 2 THEN
```

```
    DISPLAY "----- GROCERY LIST -----"
```

```
    FOR each item in the lists
```

```
        CALCULATE total_price = item_price × quantity
```

```
        DISPLAY item name
        DISPLAY item price
        DISPLAY quantity
        DISPLAY total_price
```

```
    END FOR
```

```
ELSE IF choice = 3 THEN
```

```
    SET overall_total = 0
```

```
    FOR each item in the lists
```

```
        CALCULATE total_price = item_price × quantity
        ADD total_price to overall_total
```

```
    END FOR
```

```
    DISPLAY "----- TOTAL COST -----"
```

```
        DISPLAY overall_total

ELSE IF choice = 4 THEN

    SORT item_names, item_prices, and item_quantities
    BASED ON item_prices FROM HIGHEST TO LOWEST

    DISPLAY "----- SORTED BY HIGHEST PRICE -----"

    FOR each item in the sorted lists

        DISPLAY item name
        DISPLAY item price

    END FOR

ELSE IF choice = 5 THEN

    DISPLAY "Thank you for using Smart Grocery List!"
    DISPLAY "Program ended."

ELSE

    DISPLAY "Invalid choice. Please try again."

END IF

END WHILE

END
```